Sustainable and Renewable Energy

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Sustainable and Renewable Energy (A07262)

Short Title:	Sustainable & Renewable Energy		
Faculty	Science and Computing		
Department	Science		
Cluster	Environmental Science		
Author	TWOODCOCK		
Credits	5	**Level:**	Advanced

Description of Module / Aims

The module will provide students with an understanding of international and Irish energy policy, the current status and future trends in Ireland's energy requirement, and the roles of energy efficiency and renewable energy in securing energy supply. The module will focus on illustrating opportunities for bioenergy as an alternative farm or forest enterprise, and provide students with the skill set required to carry out basic resource assessments. The module includes several field trips to both small- and large-scale renewable energy projects (e.g. wind farm, AD plant etc.).

Programmes

`SUST-0002`	BSc (Hons) in Agricultural Science (`WD_SAGRI_B`)	4 / 8 / E
`SUST-0002`	BSc (Hons) in Land Management (in Agriculture) (`WD_SBSAG_B`)	1 / 2 / E
`SUST-0002`	BSc (Hons) in Land Management (in Forestry) (`WD_SBSFO_B`)	1 / 2 / E
`SUST-0002`	BSc (Hons) in Land Management (in Horticulture) (`WD_SBSHO_B`)	1 / 2 / E

Indicative Content

- Basics of energy, forms, units, quantification, fuels, conversion and generation
- Global, European & Irish energy supply and demand
- Sustainable energy and the environment
- International & Irish energy policies
- Energy conservation
- Renewable energy sources & technology: bioenergy, wind, solar, hydro, wave
- Bioenergy in Ireland: wood, energy crops, liquid biofuels and anaerobic digestion case studies
- Resource assessment and supply chain development
- Financial analysis of energy production from renewables

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Appraise the physical and technological principles of energy generation, in particular the range of bioenergy resources, conversion technologies and markets.
2. Assess the challenges involved in sustainable energy and the potential solutions available through conservation, efficiency and renewables.
3. Solve basic problems relating to fuel and energy quantification, conversion and financial analysis.
4. Compare the available bioenergy pathways and comment on their relative technical and financial importance in an Irish context.
5. Compare Irish energy policy with that of other countries (EU & non-EU countries), and recommend areas of improvement in Irish policy.

Learning and Teaching Methods

- Lectures: will introduce basic energy principles, the technologies and energy supply systems available and provide a basis for the case study and elective project element of the module.
- Fieldtrips: will provide examples of best operating practice in the areas of energy conservation, wind and biomass energy implementation. Case studies, based on these field trips will guide the student in identifying critical parameters in developing successful implementation of sustainable energy.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Assignment	50%	1,2,5
Project	30%	3,4
Presentation	20%	3,4

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out practical/professional tasks. Failure to meet the objectives of projects. Unable to carry out or submit independent study and research.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on practical/professional tasks. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/professional tasks. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on practical/professional tasks. Show initiative, individual thought, and enthusiasm in self study and independent learning.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Field Trips	12	
Independent Learning	87	

Essential Material

- "COFORD Wood Energy Website." www.woodenergy.ie. http://www.woodenergy.ie/
- "IEA Bioenergy." www.ieabioenergy.com. http://www.ieabioenergy.com/
- "Sustainable Energy Ireland." www.seai.ie. http://www.seai.ie/
- Assmann, D., U. Laumanns and D. Uh. _Renewable Energy: A Global Review of Technologies, Policies and Markets_. London, UK: Earthscan, 2012.
- Bechberger, M., D.T. Reiche and P. Lang. _Handbook of Renewable Energies in the European Union_. Dublin, Ireland: Earth Horizon Productions, 2005.
- SEAI. _Renewable Energy in Ireland 2012 February (2014 Report)_ by Howley, M., M. Holland and K. O'Rourke.. Dublin. 2012.

Supplementary Material

- "European Forum for Renewable Energy Resources." www.eufores.org. http://www.eufores.org/
- "Irish Bioenergy Association." www.irbea.ie. http://www.irbea.ie/